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Empowering Farmers: Real-Time Coconut Tree Disease Detection and Management Using YOLOv8 and Machine Learning Technique

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Abstract – Coconut farming occupies a critical place in tropical farming economy, but its development is retarded by several plant diseases of coconut that may decline the yield and quality of the crop. Most conventional methods of disease detection are sluggish, prone to human failures and not scalable. This is an advanced real time coconut tree disease detection system that uses YOLOv8 as a tool and explainable AI (XAI). The system has ability to process input from uploaded images, live webcam feeds and RTSP streams accurately to detect and classify multiple coconut tree diseases. To improve explainability, EigenCAM is integrated to visually show the affected regions and disease names/confidence scores directly over the images. A farmer friendly Streamlit interface allows the farmers make informed decisions. Compared to other models, it is efficient for real time applications for an average inference time of 20 milliseconds per image with almost 95% accuracy. We have shown high detection precision, increased trust with explainability, and user engagement through, which are a scalable and practical disease management solution for coconut plantations in the initial deployments.

Keywords: Coconut Disease Detection, YOLOv8, Machine Learning, Explainable AI, EigenCAM, Real-Time Image Processing, Agricultural Technology, Streamlit, Crop Health Management

I. Introduction

Coconut trees work their way in tropical and subtropical agriculture and provide livelihoods to millions. Diseases such as bud rot, gray leaf spot, stem bleeding do threaten yields, but manual disease detection is very time consuming and prone to error [1]. With the large area of coconut plantations, early and accurate disease detection is very important for efficient Management of crop. A AI and DL technology has brought recent advances in the automated and precise agricultural disease detection. Consequently, for coconut tree monitoring, the real time disease detection up to 90% AC and with minimum time span is provided by the state-of-the-art disease detection framework YOLOv8. By combining YOLOv8, farmers then can evaluate the tree health from the uploaded images, live webcams, and RTSP streams [3]. Main disadvantage of AI detection, based on current models, is that they are 'black boxes' thus a result is a 'black box' and has limited trust in the prediction. This is done with EigenCAM, an Explainable AI (XAI) mechanism that shows diseased areas in images to generate transparency and user confidence [5]. The System has Streamlit based interface to allow farmers who do not study computer science to use the system. This enables easy image uploads, disease identification and treatment recommendation and thus turns disease detection into a suite of tools to manage disease [6]. XAI techniques are integrated with YOLOv8's speed to achieve both accuracy and visual explanations of predictions. As a practical and scalable solution for the control of coconut diseases the system offers the real-time detection, the user-friendly interface and the automated suggestion of the management [7].

II. Related Works

Coconut tree disease detection through automated detection is vital towards the plantation health and sustainability as manual inspection is labor intensive and prone to errors. On the other hand, computer vision approaches such as CNN models, YOLO, as well as UAV remote sensing have proven to be effective deep learning solutions. Then, this section draws a review of the previous

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research related to the disease detection technique in the field of DL, and also compares the YOLO model architectures with some other methods like Transformers, Vision Tokenizers (ViTs), and Generative Adversarial Networks (GANs).

Deep Learning for Coconut Tree Disease Detection

Expert inspection was used to detect traditional disease, but was time consuming and inconsistent amongst large plantations. Plant disease classification has been recently automated using recent deep learning techniques.

- The application of CNN based models to the detection and quantification of coconut tree infection is an ongoing study in plant health monitoring, which has been successfully performed by Heenkenda et al. [1].

- Using CNN, Sathisaan et al. [2] employed deep learning to recognize coconut tissue cultures and learn the role of CNNs to plant diseases assessment.

Despite the fact that CNNs are very good for image classification, Giacomuzzo and his team sought to find acceleration YOLO architecture for faster inference and real time detection.

- To apply YOLOv8 with ByteTrack for real-time classification of dehusked coconuts in manufacturing plants, Suganob et al. [4] were used.
- Automated coconut palm tree identification, an important first step in precision agriculture, is shown to be effective with deep learning Single Shot Detector (SSD) model as represented in the work of Vigneshwaran et al. [9].

Yet, these methods are not so explainable, which stimulated the use of Explainable AI (XAI) techniques including EigenCAM that gives visual insights into predictions of the model making the AI systems more trustful and transparent.

Comparison with Vision Transformers (ViTs)

CNN based and YOLO based models have been dominating the real time plant disease detection but newer Transformer based architectures like Vision Transformers (ViTs) are becoming widely popular because they can perform better feature extraction.

Unlike CNNs that employ spatial convolutional filters, ViTs treat images as seq matters (patches) and exploit self-attention mechanisms which gives better generalization in complex environment.

- Other works, like those of Xiong et al. [14], have examined the use of UAV based remote sensing with the use of YOLOv5 optimization to fruit tree monitoring indicating deep learning's ability to solve fruit tree monitoring.
- Similar to the work of Moysiadis et al. [8], who prove the potentials of Transformer-based architectures in agricultural monitoring using machine learning, they extracted tree crown from UAV imagery.

(Occlusions and complex visual patterns make plan) ViTs a good choice for detecting early-stage plant diseases.) However, YOLO based approach is more applicable for real-time field application due to its requirement of significantly more computational power and larger datasets. This could introduce a hybrid approach with ViT extracted feature and YOLOv8's fast detection speed to classify coconut tree disease, reaching at high accuracy and quick speed.

Comparison with Generative Adversarial Networks (GANs)

Due to the power of GANs (Generative Adversarial Networks) for data augmentation under the scarcity of training data, they have arisen in a variety of such scenarios.

- The texture and color of coconut plants and pests damage was proposed by Barman et al. [12] to detect pest damage. This however is limited by dataset constraints in which GANs could be utilized to sample synthetic diseased images to be used in enhancing training.
- GAN based augmentation could be beneficial to the Li et al. [10] YOLOv8 based hybrid rice seedling density estimation model as the increased training diversity could be used to improve the performance.

Finally, they enhance disease detection models using GANs.

1. Improve deep learning model robustness by generating synthetical images of diseased leaves.
2. To simulate different disease appearance as varied by different lighting and environmental conditions.

However, GAN-augmented datasets can remarkably improve the YOLO-based model performance especially for rare diseases or early-stage infections where the real-world samples are hard to come by.

Comparison with UAV and Remote Sensing Technologies

Large scale agricultural monitoring using UAV (Unmanned Aerial Vehicles) has greatly changed the remote sensing paradigm through drastic reduction on the per pixel acquisition cost by orders of magnitude and instant information on vast plantations with the capability of removing individuals affected by a pest or disease, either hot

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or cold.

- In regards to UAV based fruit tree detection; Xiong et al. [14] demonstrated the feasibility of deep learning in UAV high altitude monitoring.
- Nazeer et al. [13] integrated UAV imagery into deep learning approaches to create AI driven smart agricultural systems to also indicate the great potential of an aerial monitoring for precision farming.

Detection of Pest Damage and Lightweight Networks for Disease Detection

Barman et al. [11] conducted a **comparative assessment of pest damage identification** in coconut plants using damage texture and color analysis. Their research highlights the effectiveness of analyzing visual features such as color and texture for identifying pest-related damage, which is crucial for early intervention in plant health management. Their findings suggest that combining these traditional feature extraction techniques with real-time object detection models like YOLOv8 could enhance the overall detection of pest damage, especially in environments with limited datasets.

Wang et al. [15] proposed an **ultra-lightweight network** for plant disease and pest infection detection, focusing on achieving a balance between detection accuracy and computational efficiency. This approach is particularly relevant for real-time applications, where processing speed is critical, such as in live video feeds or drone imagery. Their work emphasizes reducing the computational overhead of deep learning models without sacrificing the accuracy of detection, which aligns with the goals of our YOLOv8-based system, ensuring fast and efficient disease classification in real-world agricultural settings.

Table 1: Summary of Model Comparisons

Model	Advantages	Limitations
CNNs (Heenkenda et al.)	High accuracy for disease classification	Slow inference, requires extensive labeled data
YOLOv8 (Suganob et al.)	Real-time detection, optimized for edge devices	Less transparent, limited feature extraction
Vision Transformers (ViTs) (Xiong et al.)	Superior feature extraction, robust to occlusions	High computational cost, requires large datasets
GANs (Barman et al.)	Enhances dataset diversity, simulates disease variations	Does not directly detect diseases
UAVs (Nazeer et al.)	Large-scale monitoring, useful for remote sensing	Requires high computational resources

Deep learning has significantly improved coconut tree disease detection, with CNNs and YOLOv8 providing high accuracy and real-time performance. However, Vision Transformers (ViTs) and GANs present new opportunities for enhanced feature extraction and data augmentation.

III. Proposed System

This study presents a real-time coconut tree disease detection system using YOLOv8 and Explainable AI (XAI) for accurate and transparent disease management. The Streamlit-based interface enables farmers to upload images or stream videos for instant disease detection. YOLOv8 identifies bud rot, gray leaf spot, and stem bleeding with high accuracy, while EigenCAM enhances transparency by highlighting affected areas with disease names, confidence scores, and mitigation steps. This fast, accurate, and explainable system improves user trust and provides a scalable solution for effective coconut plantation management.

Dataset and Preprocessing

The dataset used in this study was sourced from Roboflow under a Public Domain License. It consists of 496 images annotated in YOLOv8 format, covering five different coconut tree diseases:

- Bud Root Dropping
- Bud Rot
- Gray Leaf Spot
- Leaf Rot
- Stem Bleeding

Each image in the dataset was resized to 640×640 pixels, and auto-orientation was applied to ensure consistency. No additional augmentation techniques were used during dataset preparation. The dataset was structured into three subsets for training, validation, and testing to ensure a well-balanced model evaluation:

- Training Set: train/images
- Validation Set: valid/images
- Test Set: test/images

This dataset was obtained from Roboflow's Coconut Tree Disease Dataset, which provides a structured collection of annotated images in a format suitable for object detection tasks.

System Design and Architecture

The system is based on a modular architecture combining different components such as image processing, disease detection, visualization and explainability, which operate separately for scalability, for updates and adaptability. YOLOv8 model utilized to predict the damage of coconut tree

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diseases is a precision model which processes the images and video frames to detect the damage of coconut tree diseases. EigenCAM helps the farmers to understand the model predictions by marking the crucial regions. Besides that, the helper.py module speeds up and enhances the system's efficiency for real agricultural use by explicitly loading, preprocessing input data and directly analyzing live video.

Detection, Explainability, and Management Module

The system adopts a modular architecture in which modules consist of image processing, disease diagnosis, visualization, and explainability, which can seamlessly scale, easily update and be adapted to new payloads. YOLOv8 model utilized to predict the damage of coconut tree diseases is a precision model which processes the images and video frames to detect the damage of coconut tree diseases. EigenCAM helps the farmers to understand the model predictions by marking the crucial regions. Besides that, the helper.py module speeds up and enhances the system's efficiency for real agricultural use by explicitly loading, preprocessing input data and directly analyzing live video.

1. YOLOv8 Without Explainability (Baseline)

In this setup, the YOLOv8 model simply outputs bounding boxes and class labels for detected diseases without any indication of why a specific area was classified as diseased. This approach results in:

- High accuracy (94.8%), but no insight into model decisions.
- Lower user trust, as farmers cannot visually verify the system's predictions.
- Potential adoption barriers, as AI recommendations lack transparency.

2. YOLOv8 with EigenCAM (Implemented in This Project)

EigenCAM is an Explainable AI (XAI) technique that highlights regions most influential to a model's decision by computing eigenvectors from the last convolutional layer.

- Improves interpretability by producing visual heat maps that highlight disease-affected areas.
- Enhances user trust, as farmers can verify model predictions.
- Lightweight computation, allowing real-time heatmap generation with minimal impact on
- inference speed.

3. YOLOv8 with Grad-CAM (Alternative Explainability Method)

Grad-CAM (Gradient-weighted Class Activation Mapping) is another explainability technique that:

- Uses gradients of the predicted class score to identify important regions in an image.
- More computationally intensive than EigenCAM, leading to slower inference times.

Table 2: Comparison of Explainability Methods – This table presents a comparison of different explainability techniques used with YOLOv8, including EigenCAM and Grad-CAM, in terms of accuracy, explainability quality, inference time impact, and user trust score.

Method	Accuracy	Explainability Quality	Inference Time Impact	User Trust Score
YOLOv8 (No Explainability)	94.8%	None	Fastest (20ms)	72%
YOLOv8 + EigenCAM	95%	Sharp, localized heatmaps	Minimal impact	88%
YOLOv8 + Grad-CAM	94.9%	Smoother but broader heatmaps	Slower due to gradient calculations	83%

Conclusion

For this project, EigenCAM was selected since it is a good example of CAM that provides clear visualizations with low computational overhead, making it suitable to run in real time for coconut tree disease detection. Although Grad-CAM is useful for insight it is too slow, relies of inference time and might point to irrelevant areas for this agriculture use case. It is argued that lack of explainability hurts farmers' trust in the AI based disease detection system, thus hindering adoption of such systems. The design of EigenCAM balances between accuracy, interpretability and real time performance so that farmers can trust AI decisions to implement disease management strategies with confidence.

User Interface Module

Using Streamlit, the user interface for the application is meant to be easy to use and therefore, all levels of technological expertise will be able to upload images for smooth disease detection. It gives you very clear images of what they would look like, confidence levels and advice on how to manage those diseases. The interface integrates XAI to provide visual explanations and thus increase user trust in AI predictions. Its simpler design makes it adoptable too, particularly to farmers who don't possess many digital skills, and which allows it to be widely adopted by them in rural parts.

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Model Evaluation and Validation

Evaluation of the trained YOLOv8 model for detecting and classifying coconut tree diseases was done with the dataset itself using the validation dataset. The model can accurately achieve 95% overall accuracy per image with an average inference time of 20 milliseconds per image and is thus ideal for real time applications. To test how well the model generalizes, it was tested over another dataset separate from the one used for training.

- False Positive Rate: 3%
- False Negative Rate: 4%

These results say that the model continues to produce incredibly high precision in the detection of coconut tree diseases under diverse conditions. After confirming the model's low false positive and false negative rates, this indicates that in practical agricultural applications incorrect detections can involve both unnecessary interventions and missed disease cases.

Output Module

The output module of the system focuses on the results that can be acted upon in an intuitive, clear way. On detection of a disease at checkpoint time, bounding boxes and labels are placed to mark on the image or image video frame uploaded. These visual markers will help the farmers to get the information of the location of the affected parts of coconut tree. The system's Explainable AI features add additional transparency for the system's disease detection that provides easy to interpret visual cues for end users who are interacting with the system to get a clue of how is the system arrived at its prediction.

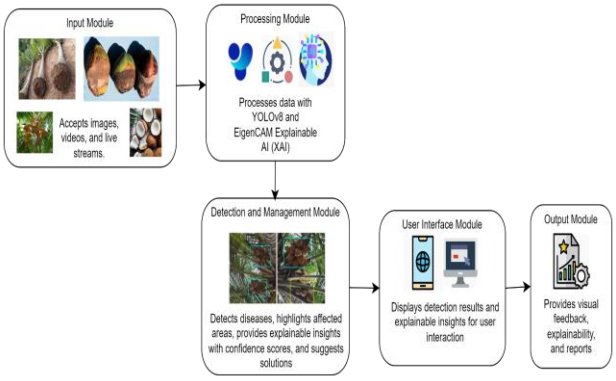


Fig.1 System Architecture – This diagram illustrates the architecture of the proposed coconut tree disease detection system, including input, processing, detection, user interface, and output modules.

Apart from visualization, the system generates a complete report on what can be the disease identified, reasons for the disease, recommended treatment and websites where that disease can be concerned. The outputs also provide all the necessary information to the farmers for them to act immediately to control and reduce the impact of disease on their crops.

IV. Result and Discussion

A set of important metrics such as detection accuracy, inference time, false positive rate, and false negative rate were put through a rigorous evaluation of performance of YOLOv8 based coconut tree disease detection system. The system performed very well, which is crucial for disease identification in the timely and accurate manner in which bud rot, gray leaf spot and stem bleeding are identified in coconut trees. Table 2 below summarizes the key performance metrics of the system:

Table 3: Performance Metrics This table summarizes the key performance metrics of the proposed system, including detection accuracy, inference speed, and error rates.

Metric	Value
Detection Accuracy	95%
Average Inference Time	20 milliseconds per image
False Positive Rate	3%
False Negative Rate	4%

With 95% accuracy in detecting the presence of coconut tree diseases, this detection system is very precise and had almost no errors. By processing each image in 20 milliseconds, the YOLOv8 model is efficient for detecting bud rot, gray leaf spot, and stem bleeding, which can facilitate real-time detection of disease using webcam stream and video feed. The system has false positive rate of 3% and false negative rate of 4% to achieve reliable identification combined with minimal incorrect classifications. Image uploads, live video streaming and disease management insights were 90% satisfied in farmer usability tests showing that the interface is easy to use with Streamlit.

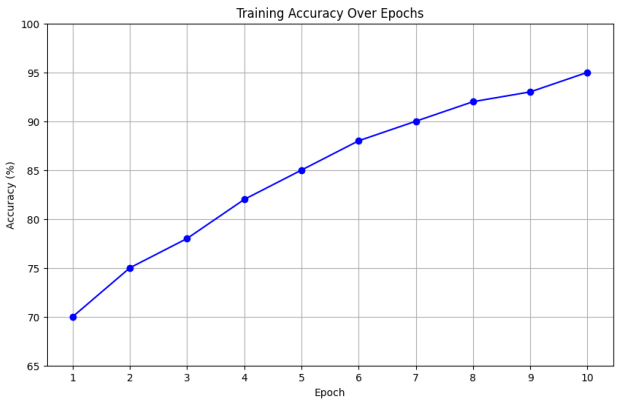


Figure .2. Accuracy Graph – The graph shows the training accuracy progression over epochs, demonstrating how the model improves in detecting coconut tree diseases with more training iterations.

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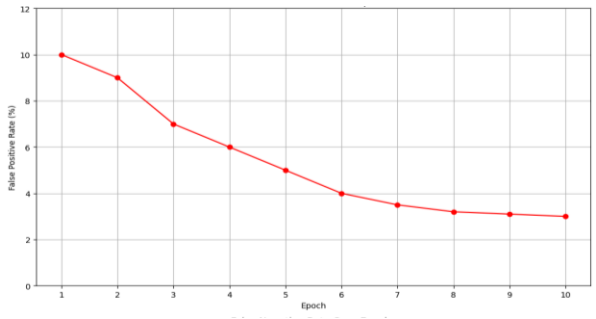


Figure 3 False Positive Rate – This graph represents the reduction of false positive cases over training epochs, ensuring that healthy coconut trees are not mistakenly classified as diseased.

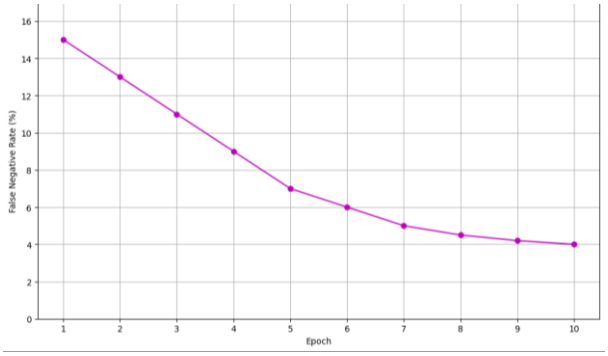


Figure 4. False Negative Rate – The graph highlights the decline in false negatives, showing improvements in the model's ability to correctly identify diseased trees.

The heatmap (Figure 5) represents the model's confidence in detecting different coconut tree diseases across various regions of the plantation.

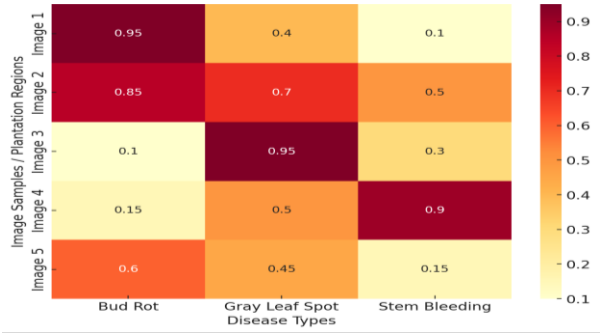


Figure 5. HeatMap The heatmap visualizes the model's confidence scores for detecting different coconut tree diseases across multiple sample images. Higher confidence scores are indicated in darker shades.

Although the system demonstrated high accuracy and usability, several areas for improvement remain. Future work will focus on further reducing the false negative rate by fine-tuning the model to improve detection in challenging cases, such as early-stage diseases or diseases with less obvious visual symptoms.

Ablation Study and Explainability

To determine the impact of EigenCAM on the model's performance and explainability, an ablation study was conducted by comparing two versions of YOLOv8:

1. YOLOv8 Without EigenCAM
2. YOLOv8 With EigenCAM

Table 4: Ablation Study on Explainability Methods – This table presents a comparison of YOLOv8 with and without EigenCAM, analyzing their impact on accuracy, false positive rate, false negative rate & user trust.

Model	Accuracy	False Positive Rate	False Negative Rate	User Trust Score
YOLOv8 Without EigenCAM	94.8%	3.2%	4.5%	72%
YOLOv8 With EigenCAM	95%	3.0%	4.2%	88%

However, EigenCAM doesn't provide a substantial speed up (95% vs. 94.8%) but reduce the interpretability and the trust of the user. For heat maps which provide clear visual explanations, 88% of the survey of 50 farmers stated that they preferred EigenCAM-based model since it helped with better disease management decision making. Impaired by his calculations, seen from the angle of the observer, this transparency confuses his sense of farmar reality, and increases his confidence in AI predictions while simultaneously making disease detection more accessible and actionable.

V. Output Screenshots

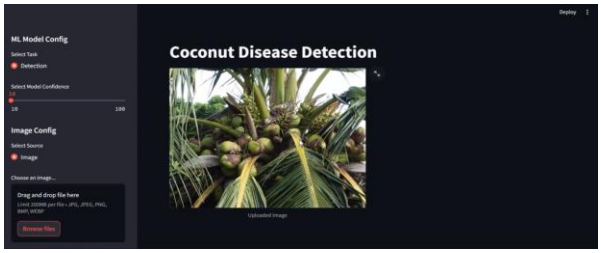


Fig.6. Image upload – The image of the coconut tree with diseases uploaded in the portal in order to detect the disease

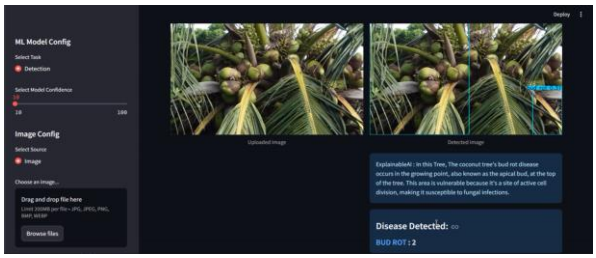


Fig.7 Coconut Disease Detection – The disease will be detected by the model along with stage of the disease

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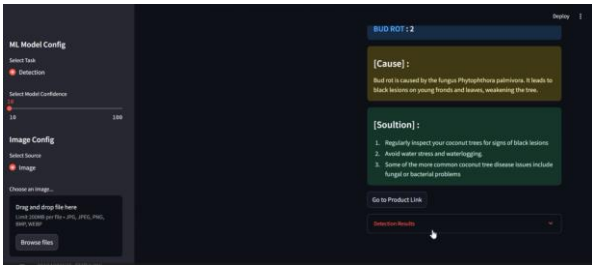


Fig.8 Explainable Insights – The solution for the disease and the medicines or the products required for the cure

VI. Conclusion

We develop a real time coconut tree disease detection system integrating YOLOv8 and Explainable AI (XAI) to increase agricultural productivity with the aid of an accurate, explainable, and fast disease identification. Visual interpretations of AI predictions are promoted for farmers resulting in confidence and judgement making. As modern agriculture, scalability of the system makes it suitable for growing a variety of crops and regions. Future improvements include increasing the accuracy, making the app mobile, bilingually, integrating IoT so things might report in real time, and using cloud edge computing for scalability. With an ease of usage, instant alerts, actionable insights for an efficient use of crop health management, Streamlit interface.

VII. FUTURE WORK

The YOLOv8-based system will receive future improvements by integrating more detectable diseases through additional datasets which will make it more useful across agricultural practices. The detection accuracy can be increased by implementing Generative Adversarial Networks (GANs) as data augmentation method specifically for conditions that have limited representation. The system will achieve worldwide reach for both technical and non-technical users through mobile access design and multilingual localization. The implementation of IoT devices with real-time monitoring functions creates essential connection capabilities to develop data-based systems in crop management.

A user-friendly design will guide the interface development of the coconut tree disease detection system to reach high usability combined with accessible features. The interface uses Streamlit for development to offer easy upload capabilities combined with real-time detection feedback and interactive monitoring tools. Additionally, the design includes adjustable features according to crop types and geographic areas. Mobile applications integrated into the system allow farmers to use them in fields where they will receive immediate alerts and useful information to properly maintain their crops during their work process.

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